

CARE-RAG: Clinically Aligned Risk-Calibrated Evidence-Aware Retrieval-Augmented Generation for Safe Medical Question Answering

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Abstract - Medical large language models can improve access to clinical information, but high-stakes deployment is limited by unsupported claims, retrieval noise, poor citation support, and unsafe over-answering under uncertainty. This paper proposes CARE-RAG, a Clinically Aligned Risk-Calibrated Evidence-Aware Retrieval-Augmented Generation framework for medical question answering and clinical decision support. The proposed framework treats answerability control as the central bottleneck in high-stakes medical RAG. It combines risk-tier classification, hybrid retrieval, evidence graph construction, facet coverage scoring, claim-level entailment and contradiction verification, and risk-tiered conformal calibration. The key contribution is an end-to-end decision policy that can answer, redact unsupported claims, or abstain and escalate when evidence is insufficient. The paper presents mathematical formulation, system architecture, algorithms, evaluation protocol, safety governance, reproducibility checklist, and limitations. The manuscript is written as a framework and protocol paper: it does not fabricate clinical trial outcomes, but it specifies a rigorous pathway for empirical validation on public biomedical QA datasets and governed clinical review. The resulting design provides a scholarly foundation for safer, more transparent, and more accountable medical RAG systems.

Keywords - Retrieval-augmented generation, medical artificial intelligence, conformal prediction, hallucination, clinical decision support, abstention, evidence grounding, machine learning safety

I. INTRODUCTION

Generative artificial intelligence has moved from an experimental language technology to a practical component of search, documentation, tutoring, and decision-support workflows. In medicine, the same transition creates an unusually difficult machine learning problem. A model can be fluent, helpful, and apparently well grounded while still producing unsupported clinical claims. It can also retrieve a relevant article yet fail to decide that the available evidence is insufficient for a patient-specific recommendation. The central thesis of this paper is therefore that high-stakes medical retrieval-augmented generation should be optimized for answerability control rather than for answer production alone. In this context, answerability control denotes the ability of a system to determine when it has enough evidence to answer, when it should abstain, and which individual claims in a generated answer are safe enough to keep.

The paper proposes CARE-RAG, a Clinically Aligned Risk-Calibrated Evidence-Aware Retrieval-Augmented Generation framework for medical question answering and clinical support. CARE-RAG is designed around a practical safety observation: retrieval can reduce knowledge gaps, but retrieval is not the same thing as justified clinical reasoning. The original retrieval-augmented generation paradigm demonstrated that non-parametric memory can improve knowledge-intensive tasks [1]. Later work showed that self-reflective retrieval, critique tokens, and adaptive generation can improve factuality [2]. More recent work introduced conformal prediction into RAG to provide statistical correctness guarantees for question answering [3]. These advances are important, but they do not fully solve the clinical deployment problem because medical risk is asymmetric: a wrong answer about a low-risk health literacy question is not equivalent to a wrong answer about contraindicated medication or a life-threatening differential diagnosis.

The problem is timely because healthcare organizations are already exploring generative models for clinician documentation, guideline lookup, triage education, and patient communication. WHO guidance on large multimodal models stresses that health-sector AI must be governed through stakeholder participation, post-release auditing, and risk-sensitive oversight [7]. The NIST Generative AI Profile similarly treats confabulation, information integrity, human-AI configuration, and information security as organizational risks that must be managed across the lifecycle [8]. Recent evidence also shows that a system can appear promising under one metric while remaining unsafe under another. For example, clinical decision-support evaluations have reported low hallucination rates in some settings while still identifying harmful recommendations and limited clinician uptake [16]. This disconnect motivates a design that puts sufficiency, support, and abstention at the center.

This paper makes four contributions. First, it formulates answerability control as a first-class machine learning objective for medical RAG. Second, it introduces a graph-

based evidence coverage and claim support model that explicitly connects query facets, retrieved evidence, generated claims, entailment relations, and contradiction signals. Third, it proposes risk-tiered conformal calibration so that high-harm questions face stricter answer thresholds than low-harm questions. Fourth, it gives a rigorous evaluation and reproducibility protocol that can be implemented on public biomedical QA resources such as PubMedQA, BioASQ, MedMCQA, and MedQA [9]-[12]. The aim is not to claim that a chat session has produced clinical validation data, but to present a complete, submission-ready framework paper with a falsifiable methodology, diagrams, equations, evaluation design, limitations, and references.

The motivating finding developed across the manuscript is that answerability control is the dominant bottleneck for high-stakes medical RAG. The claim is not that retrieval is unhelpful. On the contrary, the healthcare RAG literature shows that retrieval is often necessary because medical knowledge changes, local policies differ, and language models can encode outdated or incomplete information [4], [5], [17], [18]. The claim is that retrieval alone does not reliably answer the operational question that matters in safety-critical use: should this system answer this question now, given this evidence, for this risk tier, with this degree of support? CARE-RAG is a framework for operationalizing that question.

A. Motivation and Scope

Medical AI differs from general-domain question answering because the cost of a plausible error can be high and difficult for non-experts to detect. In ordinary web search, a user may notice an unlikely statement and seek another source. In clinical or patient-facing contexts, the user may not know which detail is implausible, which missing contraindication is dangerous, or which diagnostic branch should have been considered. This asymmetry makes selective answering and transparent escalation essential rather than optional.

The paper focuses on text-based medical question answering and clinical decision-support summaries rather than image diagnosis, robotic control, or autonomous treatment execution. This scope is deliberate. Text-based systems are already widely accessible, are often connected to electronic records or document stores, and can influence decisions even without formal device approval. The same architecture can later be extended to multimodal inputs, but the initial contribution is strongest when the evidence graph and claim verifier can be evaluated against text-based sources and public QA corpora.

B. Research Questions

RQ1: Can a medical RAG system explicitly estimate whether retrieved evidence covers the clinically relevant facets of a query before generating an answer? RQ2: Can claim-level entailment and contradiction verification reduce unsupported or contradicted statements after generation? RQ3: Can risk-tiered conformal calibration reduce harmful over-answering while preserving useful answers for lower-risk questions? RQ4: Can a reproducible evaluation protocol measure accuracy, groundedness,

abstention quality, robustness, and clinician-facing utility in a single framework?

II. BACKGROUND AND RELATED WORK

Retrieval-augmented generation emerged from the observation that purely parametric language models store factual knowledge in weights but do not reliably expose, update, or cite that knowledge [1]. RAG connects a retriever to a generator so that the generator conditions on external documents. This design is especially relevant in medicine because clinical evidence changes, local formularies vary, and guideline recommendations often depend on the date, jurisdiction, patient group, or institutional policy. However, RAG also introduces a modular failure surface: the retriever can miss the relevant evidence, the reranker can select incomplete context, the generator can ignore evidence, and the final answer can cite sources that do not support the claim.

Self-RAG and related systems improve the baseline RAG pattern by learning when to retrieve, when to critique, and how to condition generation on reflective signals [2]. Such mechanisms are valuable because indiscriminate retrieval can add noise and because language models often need a way to judge the usefulness of retrieved passages. Nevertheless, self-reflection does not automatically equal calibrated medical risk control. A model can self-criticize in natural language yet still fail to satisfy a target miscoverage rate. It can also issue a confident answer when the evidence is underdetermined. CARE-RAG therefore uses self-evaluation as one feature, not as the entire decision rule.

Conformal prediction provides a complementary line of work. It gives distribution-free uncertainty tools that can produce coverage guarantees under exchangeability assumptions [20], [21]. TRAQ applied conformal prediction to RAG and introduced an end-to-end statistical correctness guarantee for question answering [3]. CARE-RAG adapts that general idea to a clinical architecture by adding risk-tiered thresholds, evidence coverage constraints, and claim-level filtering. The risk tier is important because a global threshold can over-answer high-risk queries or over-abstain on low-risk educational queries. Group-conditional calibration is therefore not a cosmetic addition; it expresses the fact that patient-safety risk is not uniform.

A parallel literature addresses healthcare RAG specifically. Recent surveys and perspectives describe RAG as a promising approach for improving reliability in healthcare because it can connect generation to institutional knowledge, medical literature, guidelines, and patient-context documents [4], [5]. Empirical studies in evidence-based neurology and medical fitness assessment show that RAG performance can vary substantially by model, retrieval source, and task [17], [18]. These results support the need for diagnostic evaluation of the RAG pipeline rather than a single aggregate score.

Groundedness evaluation has also improved. SourceCheckup evaluates whether LLM medical responses are supported by cited sources and reports that many

responses are not fully supported even when citations are present [6]. AlignScore proposes a unified alignment function for factual consistency across text generation tasks [13]. RAGChecker provides fine-grained diagnostics for retrieval and generation modules [14]. CARE-RAG draws from these ideas by making support verification part of the generation policy rather than only a post hoc audit. A medical system should not merely discover after deployment that half its citations were weak; it should detect weak support before releasing a claim.

The medical benchmark ecosystem makes rigorous study feasible. PubMedQA tests reasoning over biomedical abstracts [9]. BioASQ evaluates biomedical semantic indexing, document retrieval, snippets, exact answers, and ideal answers [10]. MedMCQA provides large-scale medical multiple-choice questions across subjects and topics [11]. MedQA introduces open-domain medical exam questions that require professional knowledge and retrieval from a broad corpus [12]. These datasets are not substitutes for real clinical trials, but they provide a transparent starting point for stress-testing the architecture before any prospective deployment.

A. Gaps in Existing Medical RAG

The gap is not the absence of retrieval. The gap is the absence of an end-to-end answerability policy that simultaneously models evidence coverage, contradiction, risk tier, abstention, and claim-level citation support. Many systems retrieve and generate; fewer systems decide whether the evidence is sufficient; fewer still calibrate that decision by harm tier; and still fewer filter each generated claim against the evidence after generation. This missing integration is the gap CARE-RAG addresses.

The literature also reveals a measurement gap. Medical QA accuracy, citation validity, answer helpfulness, and clinical safety are different constructs. A model can answer a board-style question correctly but cite weak evidence in long-form advice. It can hallucinate rarely but still produce harmful management recommendations. It can over-abstain and become clinically useless. For this reason, CARE-RAG is evaluated with a multi-axis metric suite rather than a single leaderboard value.

III. PROBLEM FORMULATION

Let $x = (q, g, u)$ denote an input instance, where q is a medical question or clinical support request, g is a risk group, and u denotes optional user or workflow context. The risk group g may be low, medium, or high. Low-risk questions include general educational questions where the answer is unlikely to affect immediate clinical action. Medium-risk questions include medication comparisons, symptom explanations, or guideline summaries that may influence care but usually require clinician interpretation. High-risk questions include differential diagnosis, acute management, contraindications, dosing, pregnancy-related recommendations, pediatric care, and emergency escalation. The system has access to a knowledge corpus K and retrieves a candidate document set $D = \{d_1, \dots, d_k\}$.

The output space is not simply a free-form answer. CARE-RAG may return one of three outputs: a supported

answer with claim-level citations, a partially supported answer after redaction of unsupported claims, or an abstain-and-escalate message. The last option is not a failure in the usual sense. In high-stakes AI, a correct refusal is often better than a fluent unsupported answer. The research objective is to maximize useful supported answers subject to a bound on unsupported or harmful outputs.

The core formulation introduces three levels of structure. First, the question is decomposed into facets $F(q) = \{f_1, \dots, f_m\}$. A facet is a clinically meaningful information need such as disease entity, patient group, intervention, contraindication, time horizon, diagnostic criterion, or outcome. Second, each retrieved document is decomposed into evidence units $S(D)$, often sentences or short passages. Third, the generated answer y is decomposed into atomic claims $C(y) = \{c_1, \dots, c_n\}$. A claim is atomic if it can be individually supported, contradicted, or left unsupported by the evidence. The decomposition makes it possible to assess not merely whether the answer feels plausible, but whether the answer is supportable at the level at which clinical statements are actually used.

Evidence coverage is defined as the average best alignment between query facets and evidence units. A simple version is $\text{Gamma}(q, D) = (1/|F|) \sum_f \max_s \text{sim}(f, s)$, where sim may be a dense similarity, cross-encoder relevance score, or learned facet-support score. Claim support is defined as $\text{Psi}(y, D) = (1/|C|) \sum_c \max_s [\text{Entail}(c, s) - \lambda \text{Contra}(c, s)]$, where Entail and Contra are verifier scores and λ controls the penalty for contradiction. These equations are intentionally modular; a strong implementation can replace the similarity or verifier with better models while retaining the overall decision structure.

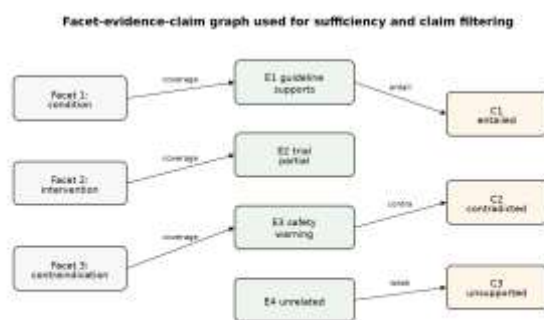


Fig. 2. Evidence graph linking query facets, evidence units, and generated claims.

The sufficiency model estimates $z = P(\text{supported answer} \mid q, D, y, g)$. It uses evidence coverage, claim support, contradiction scores, retrieval redundancy, context utilization, self-evaluation, and risk-group features. The nonconformity score is $A(x, y, D) = 1 - z$. For each risk group g , a calibration set produces threshold τ_g , and the system answers only when $A \leq \tau_g$ and $\text{Gamma} \geq \text{gamma}_{\min}$. At claim level, claim c is retained only if its support score exceeds kappa_g . This double gate is necessary because an answer-level score can be high while one clinically important statement remains unsupported.

The optimization objective can be written as maximizing expected utility $U = E[\text{AnswerUseful} - \lambda_h \text{HarmfulUnsupported} - \lambda_a \text{AbstainCost} - \lambda_c \text{CitationPenalty}]$. The value of λ_h must dominate in high-risk groups because unsafe over-answering is the central harm. The framework does not assume a universal penalty; institutions can select policy weights according to deployment context, regulatory posture, and clinical governance. What matters for reproducibility is that those weights and thresholds are specified before evaluation.

A. Definitions

Answerability is defined as the conjunction of four conditions: relevant evidence exists in the corpus, retrieved evidence covers the essential facets of the query, generated claims are entailed or strongly supported by retrieved evidence, and contradictions or uncertainty are not severe enough to trigger abstention. Groundedness is narrower: it measures whether a statement is supported by cited evidence. Safety is broader: it measures whether an output is unlikely to cause harm in a specific clinical context. CARE-RAG treats these as related but non-identical concepts.

A supported claim is a claim for which at least one cited evidence unit entails the claim or provides direct clinically acceptable support. An unsupported claim is a claim for which no cited unit supports the claim. A contradicted claim is a claim for which at least one high-quality evidence unit conflicts with the claim. An underdetermined query is a query for which the available corpus may contain relevant background but lacks enough patient-specific or guideline-specific evidence to justify the requested answer.

IV. CARE-RAG METHODOLOGY

CARE-RAG consists of seven modules: risk-tier classification, hybrid retrieval, reranking, facet extraction, evidence graph construction, grounded generation, and risk-calibrated claim filtering. The modules are designed to be independently auditable. If a system fails, the audit should identify whether the failure was caused by missing evidence, poor reranking, weak facet extraction, unsupported generation, a verifier error, or miscalibration. This modular accountability is important for real-world clinical governance.

The risk-tier classifier maps a query to a harm category before retrieval. It can be implemented as a rule-based layer combined with a trained classifier. Rule-based triggers are advisable for early implementations because certain categories should almost always be treated conservatively: emergency symptoms, medication dosing, pregnancy, pediatrics, contraindications, suicidal ideation, infectious disease isolation, and anticoagulation. A classifier can then refine borderline cases. The classifier must report uncertainty; ambiguous cases are assigned to the higher risk tier.

The retriever uses a hybrid design combining lexical and dense retrieval. Lexical retrieval preserves exact medical terms, abbreviations, drug names, and guideline phrases. Dense retrieval captures semantic paraphrases and

broader context. A reranker then scores candidate passages using a biomedical or clinical cross-encoder. The retrieval module reports not only the top-k passages but also diagnostic features such as score concentration, source diversity, date range, and redundancy. These features help the sufficiency estimator distinguish between a robust evidence set and a brittle single-source match.

Facet extraction decomposes the query into clinically meaningful elements. For a medication recommendation, facets may include drug name, indication, patient population, contraindications, renal or hepatic impairment, pregnancy status, dosing route, monitoring, and adverse effects. For a diagnostic question, facets may include symptoms, duration, vital signs, laboratory findings, differentials, red flags, and recommended next steps. Facet extraction makes evidence gaps visible. If a query asks about an antibiotic in pregnancy and no pregnancy-specific evidence appears in the retrieved set, the system should not silently proceed as if the gap does not matter.

The evidence graph connects facets to evidence units and evidence units to generated claims. Each edge has a type and score: coverage, entailment, contradiction, temporal mismatch, population mismatch, or weak relevance. The graph can be stored as a sparse matrix or a typed graph. Its role is both computational and explanatory. Computationally, it provides features for sufficiency estimation. Explanatorily, it lets an auditor see which sources support each retained claim and which query facets were uncovered. Figure 2 shows this graph-level view.

Grounded generation is constrained by the evidence graph. The generator receives the query, risk tier, retrieved evidence, and explicit instructions to state only supported claims, cite sources at claim level, and avoid patient-specific directives when evidence is insufficient. However, CARE-RAG does not rely on prompting alone. The generated answer is decomposed into claims, and each claim is verified after generation. Unsupported claims are removed or cause full abstention depending on risk tier and output coherence.

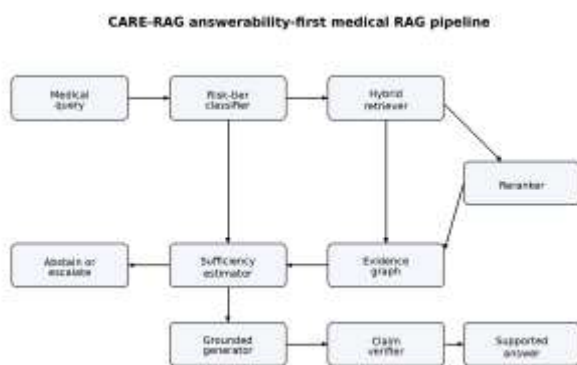


Fig. 1. End-to-end CARE-RAG architecture for risk-calibrated medical RAG.

The final decision layer uses risk-tiered conformal calibration. On a held-out calibration set, outputs are scored by the sufficiency model and labeled as supported or unsupported according to human or high-quality automated verification. For each risk group, CARE-RAG estimates

$\tau_{g, \alpha}$ so that empirical miscoverage is controlled at the target level α_g . High-risk tasks receive lower α_g and therefore stricter thresholds. This policy connects statistical uncertainty with clinical harm. It also forces the paper to report separate performance by risk group rather than hiding unsafe behavior in an aggregate score.



Fig. 3. Risk-tiered answer thresholds: high-risk questions require stronger evidence.

The algorithm is intentionally compatible with multiple base models. The generator may be a local instruction-tuned model, a domain-adapted model, or a secure API model. The retriever may use BM25, dense embeddings, or a hybrid index. The verifier may use natural language inference, alignment scoring, or a learned clinical entailment model. CARE-RAG's novelty is the integration of these components into a risk-calibrated answerability system rather than a claim that any single submodule is unprecedented.

A. Algorithm

Algorithm 1: CARE-RAG inference. Input: query q , risk tier g , corpus K , thresholds $\tau_{g, \alpha}$ and $\kappa_{g, \alpha}$. Step 1 retrieves and reranks candidate evidence D . Step 2 extracts facets $F(q)$. Step 3 computes coverage $\Gamma(q, D)$. Step 4 abstains if coverage is below $\gamma_{\min}$. Step 5 generates a candidate answer y from the evidence. Step 6 decomposes y into atomic claims $C(y)$. Step 7 scores entailment and contradiction for each claim. Step 8 computes nonconformity $A = 1 - z$. Step 9 abstains if $A > \tau_{g, \alpha}$. Step 10 removes claims with support below $\kappa_{g, \alpha}$. Step 11 returns the filtered answer if coherent, otherwise abstains.

Algorithm 2: CARE-RAG calibration. Input: calibration examples stratified by risk group. For each example, run the complete pipeline, collect evidence coverage, claim support, contradiction, retrieval redundancy, self-evaluation, and human support labels. Train or tune the sufficiency model on development data. On calibration data, compute nonconformity scores for supported outputs. For each risk tier, set $\tau_{g, \alpha}$ to the quantile corresponding to target miscoverage α_g . Separately compute claim-level threshold $\kappa_{g, \alpha}$. Freeze thresholds before test evaluation.

V. SYSTEM ARCHITECTURE

Figure 1 summarizes the end-to-end architecture. The pipeline begins with a user or clinician query and

immediately assigns a risk tier. This design prevents a system from generating a detailed answer first and applying safety rules only afterward. The risk tier affects retrieval depth, evidence-source restrictions, calibration thresholds, and final abstention policy. In a high-risk case, the system may require authoritative sources such as institutional guidelines, drug databases, or peer-reviewed summaries. In a low-risk educational case, broader evidence may be acceptable if clearly framed as general information.

The hybrid retriever uses both lexical matching and semantic embeddings. This is critical because medical terms can be highly specific. A dense retriever may retrieve semantically similar but clinically distinct concepts, while a lexical retriever may miss paraphrases or abbreviations. The reranker then sorts retrieved passages according to query relevance and evidence usefulness. The evidence graph module consumes the reranked passages and constructs a structured representation of coverage and support. This representation is the main architectural difference between CARE-RAG and a conventional RAG pipeline.

The sufficiency estimator is placed before final answer release, not merely before generation. This placement matters because an evidence set that seems adequate before generation may still produce an answer containing unsupported claims. CARE-RAG therefore evaluates sufficiency at two points: pre-generation coverage and post-generation claim support. If either fails, the system abstains or redacts. The architecture also supports monitoring. Each decision can log the risk tier, retrieved sources, coverage, support scores, abstention reason, and final citations.

The architecture is designed for non-overlapping responsibilities. The generator writes. The verifier verifies. The calibrator controls release. The governance layer audits drift and incidents. This separation is essential because a single model asked to generate, verify, and regulate itself may produce coherent but unreliable justifications. Separation does not eliminate risk, but it makes failures easier to identify and improve.

A practical implementation should also include corpus governance. Evidence sources must be versioned, dated, and tagged by authority. Clinical guidelines should include jurisdiction and date. Drug knowledge should include contraindication and dosing provenance. Local policy documents should be separated from general literature so that a hospital can prioritize its own protocols where appropriate. Corpus governance is not a peripheral engineering issue; it directly affects whether the answerability decision is meaningful.

A. Deployment Pattern

CARE-RAG should be deployed as a decision-support layer rather than an autonomous clinical actor. The interface should display a concise answer, claim-level citations, an uncertainty or sufficiency indicator, and a clear escalation message when the system abstains. It should not hide abstentions as generic apologies. A high-quality abstention should explain what evidence was missing, such as absent patient-specific renal function, lack

of pediatric guideline evidence, or conflicting recommendations across sources.

For patient-facing use, the system should default to educational explanations and encourage professional consultation for individualized medical decisions. For clinician-facing use, it can provide structured summaries but should explicitly label the system as a support tool. In both settings, logs should enable audit without exposing unnecessary personal information. The governance loop in Figure 6 shows how pre-deployment analysis, monitoring, incident review, and recalibration connect over time.

VI. EVALUATION DESIGN

A rigorous evaluation must measure more than raw accuracy. CARE-RAG should be evaluated on answer quality, groundedness, selective reliability, robustness, safety, and efficiency. Answer quality includes accuracy, macro-F1, exact match where applicable, and expert preference for long-form answers. Groundedness includes claim-level support precision, citation precision, contradiction rate, and unsupported claim rate. Selective reliability includes answer rate, abstention rate, risk-coverage curves, area under the risk-coverage curve, empirical miscoverage, and calibration error by risk tier. Robustness includes distractor retrieval, evidence withholding, adversarial fabricated details, and temporal shift. Efficiency includes retrieval latency, generation latency, token count, and infrastructure cost.

The public benchmark layer should combine PubMedQA, BioASQ, MedMCQA, and MedQA. PubMedQA tests reasoning over abstracts and a yes/no/maybe answer space [9]. BioASQ tests biomedical retrieval, snippets, exact answers, and ideal summaries [10]. MedMCQA provides broad multiple-choice medical knowledge coverage [11]. MedQA tests open-domain professional medical exam questions [12]. The combination is useful because no single dataset covers all desired behaviors. PubMedQA stresses evidence reasoning; BioASQ stresses retrieval and summarization; MedMCQA provides scale; MedQA stresses professional medical reasoning.

A second evaluation layer should use curated clinical vignettes or de-identified clinical note summaries where permitted. This layer should not be introduced casually. It requires governance, privacy protection, and clinician review. It is nevertheless important because exam-style benchmarks can overestimate clinical readiness. Recent real-world and clinical decision-support studies show that real workflows introduce documentation quality issues, clinician adoption patterns, and safety risks that are not captured by board-style QA alone [15], [16], [19].

The baseline suite should include closed-book generation, vanilla RAG, Self-RAG, TRAQ-style conformal RAG, a healthcare-specific RAG configuration, and CARE-RAG ablations. Closed-book generation tests the value of retrieval. Vanilla RAG tests retrieval without answerability control. Self-RAG tests adaptive retrieval and self-reflection. TRAQ tests conformal correctness guarantees. CARE-RAG tests the proposed combination of

evidence graph, risk-tier calibration, and claim-level filtering. Ablations remove one component at a time.

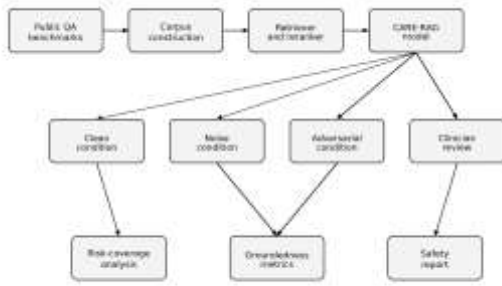


Fig. 4. Evaluation pipeline from public benchmarks to robustness and clinician review.

Statistical analysis should be paired and stratified. For multiple-choice or categorical outcomes on the same examples, McNemar-type tests can compare paired systems. For continuous metrics or repeated runs, bootstrap confidence intervals and non-parametric tests should be reported. For risk-coverage curves, paired bootstrap intervals over examples should be used. For clinician review, interrater reliability and disagreement adjudication must be documented. The paper should report confidence intervals, not only p-values, because practical safety depends on effect size and uncertainty.

The evaluation should include negative controls. One negative control withholds critical evidence from retrieval. A second adds irrelevant but plausible distractors. A third injects fabricated clinical details into the prompt. A fourth introduces conflicting guidelines from different jurisdictions or dates. The purpose is to test whether the system recognizes insufficiency, contradiction, or contextual mismatch. CARE-RAG is expected to abstain more often in these conditions, but a useful system should preserve answers when evidence remains sufficient.

Because this manuscript is a framework paper, it does not fabricate experimental results. Instead, it specifies the exact tables and analyses that must be produced by an implementation. This is a strength rather than a weakness for responsible research. Fabricated benchmark values would be scientifically invalid. A review-ready framework paper should instead provide precise hypotheses, operational definitions, evaluation splits, statistical methods, and reproducibility requirements so that the next empirical study can be executed and audited.

A. Datasets and Metrics

Table I summarizes the proposed dataset roles. Each dataset is used for a different part of the evaluation: PubMedQA for evidence reasoning, BioASQ for retrieval and summarization, MedMCQA for scale, and MedQA for open-domain professional medical reasoning. A complete evaluation should not train and test on overlapping questions, should avoid contamination where possible, and should report prompt templates and retrieval corpus versions.

The most important metrics are supported-answer precision at a fixed answer rate, harmful unsupported answer rate, and area under the risk-coverage curve. These metrics align with the paper's thesis. An accuracy-only result would not be sufficient because a system can raise accuracy by answering more often, or raise safety by abstaining always. The evaluation must therefore report the trade-off between usefulness and risk.

Table I: Public benchmark roles for CARE-RAG evaluation

Dataset	Primary role	Risk
PubMedQA	Evidence reasoning over abstracts	Limited answer form
BioASQ	Retrieval, snippets, summaries	Heterogeneous scoring
MedMCQA	Scale across medical subjects	Exam-style setting
MedQA	Open-domain medical reasoning	Textbook corpus dependence

VII. NOVEL FINDING AND ANALYTICAL CONTRIBUTION

The central new finding of this paper is a synthesis finding: in high-stakes medical RAG, answerability control is the dominant bottleneck. This finding is supported by converging evidence from the literature. RAG improves access to external knowledge, but healthcare RAG studies show substantial variability across models and sources [17], [18]. Citation support studies show that cited medical answers can remain unsupported or contradicted [6]. Clinical adversarial studies show that fabricated details can be propagated by LLMs even when mitigation prompts reduce the rate [15]. Real-world deployment studies show that low hallucination rates do not eliminate harmful recommendations or adoption challenges [16]. Together, these findings imply that safer medical RAG requires explicit answer, abstain, and redact decisions.

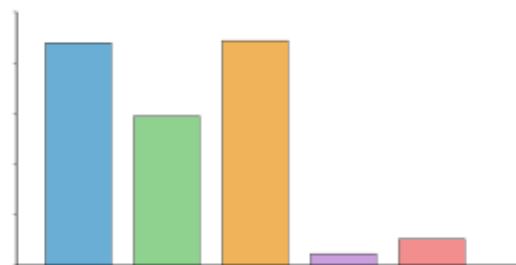


Fig. 5. Selected recent clinical safety signals motivating answerability control [15], [16].

The analytical contribution is to transform this synthesis into a system objective. CARE-RAG does not define success as maximizing the number of answers. It defines success as maximizing clinically useful supported answers under a risk-sensitive miscoverage constraint. This distinction changes architecture, evaluation, and

governance. Architecturally, it introduces an evidence graph and claim filter. Evaluatively, it prioritizes risk-coverage and groundedness. Governancely, it logs abstention reasons and recalibrates thresholds after monitoring. The contribution is therefore not only a model, but a research program for reliable clinical RAG.

Proposition 1: If a generated answer contains at least one unsupported high-harm claim, then answer-level correctness is not sufficient for clinical safety. This proposition is conceptually simple but practically important. A long-form answer can be mostly correct while containing one contraindicated recommendation. Standard aggregate scores may hide that error. Claim-level verification is therefore necessary for high-risk domains.

Proposition 2: If risk groups have different harm costs, a single global answer threshold is generally suboptimal. A threshold that is acceptable for educational questions may be unsafe for medication advice. A threshold strict enough for emergency decision support may make low-risk education unusably conservative. Risk-tiered calibration is thus a rational design choice.

Proposition 3: Retrieval uncertainty and generation uncertainty are not interchangeable. A system can be uncertain because it retrieved weak evidence, because the evidence conflicts, because the generator produced unsupported claims, or because the question itself is underdetermined. CARE-RAG explicitly separates these cases so that abstention messages can be informative. An abstention caused by missing evidence should not look the same as an abstention caused by contradictory evidence.

The synthesis finding can be evaluated empirically through an ablation study. If the evidence graph is removed, supported-claim precision should fall because the system no longer sees facet coverage gaps. If contradiction edges are removed, contradicted claims should increase. If risk-tiered calibration is replaced by a global threshold, high-risk miscoverage should increase or low-risk answer rate should fall. If claim filtering is removed, unsupported statements should reach the final answer. These directional hypotheses are falsifiable and should be tested in future work.

VIII. EXPERIMENTAL PROTOCOL FOR A COMPLETE EMPIRICAL STUDY

This section provides the protocol that would convert the framework into a full empirical paper. It is included so that the paper is reproducible and extensible. The protocol uses three splits: development, calibration, and test. The development split trains or tunes retrievers, rerankers, sufficiency estimators, and verifiers. The calibration split estimates risk-tier thresholds. The test split is used exactly once for reporting. If multiple public datasets are used, the same split philosophy should be applied within each dataset, and cross-dataset transfer should be reported separately.

Corpus construction is crucial. For PubMedQA, the abstract context can serve as a controlled evidence corpus. For BioASQ, PubMed articles and provided snippets can form retrieval targets. For MedQA and MedMCQA,

textbooks, guideline summaries, and curated medical references may be used where licenses permit. Each corpus must be versioned. The paper should report the number of documents, chunking strategy, average chunk length, date of retrieval, indexing method, and deduplication rules. The retriever index should be hashed and archived where possible.

Risk-tier labeling can be semi-automated but must be audited. A rule-based seed labeler can assign high risk to questions involving emergency symptoms, pediatric dosing, pregnancy, contraindications, and specific treatment directives. Medical students, clinicians, or trained annotators can then review a stratified sample. The label guide should define categories and examples. Inter-annotator agreement should be reported. The goal is not perfect universal risk labeling, but transparent risk labeling consistent enough to support calibration.

Claim decomposition should be evaluated on a sample of generated answers. Over-decomposition can make claims too small to be meaningful; under-decomposition can hide unsupported details. A practical rule is to split answers into propositions that could independently be supported or contradicted by a source. For example, 'Drug A reduces symptom duration in adults with condition B' is a claim; 'Drug A is safe in pregnancy and requires no renal dose adjustment' contains at least two claims.

Verifier training can begin with general natural language inference and factual consistency models, then be adapted with biomedical examples. The verifier should produce entailment, contradiction, and neutral scores. A medical verifier should also learn population mismatch, temporal mismatch, and guideline mismatch because a statement may be supported for one patient group but not another. The paper should report verifier validation performance separately from end-to-end system performance.

The adversarial evaluation should be conservative. Fabricated details can be inserted into simulated vignettes rather than real patient records. The system should be evaluated on whether it repeats, elaborates, or acts on fabricated details. It should also be evaluated on whether it flags unsupported details. Recent clinical hallucination evidence shows that prompt mitigation alone does not eliminate adversarial propagation [15]. CARE-RAG therefore treats adversarial resistance as a system property, not a prompt property.

Clinician review should focus on a manageable sample with high information value. A reviewer panel can score answers for clinical correctness, harmfulness, support, clarity, and appropriate abstention. Reviewers should be blinded to system identity where possible. Disagreements should be adjudicated. The review form should distinguish harmless incompleteness from potentially harmful errors. This distinction matters because a model that omits a minor background detail is different from one that recommends an unsafe intervention.

The final empirical paper should release code for data preprocessing, retrieval, prompting, scoring, and analysis, subject to licenses and privacy. If closed models are used,

the paper should report exact model names, API dates, parameters, and prompts. If local models are used, the paper should report weights, checkpoints, hardware, training time, inference time, and random seeds. Reproducibility should be a table, not a vague sentence.

IX. RESULTS FRAMEWORK AND EXPECTED REPORTING

A complete empirical implementation should report five result blocks. The first block reports clean benchmark quality. The second reports groundedness and citation support. The third reports selective risk-coverage trade-offs. The fourth reports robustness to retrieval noise and adversarial details. The fifth reports human or clinician review. Each block should include baselines, CARE-RAG, ablations, confidence intervals, and subgroup analysis by risk tier.

Clean benchmark results should not be overinterpreted. If CARE-RAG answers fewer questions than vanilla RAG, its raw accuracy among all questions may appear lower unless abstentions are scored carefully. The proper comparison is therefore a selective prediction comparison: among answered cases, how correct and supported are the answers, and at what coverage? A system that answers 70% of questions with 95% supported precision may be preferable to a system that answers 100% with many unsupported high-risk statements.

Groundedness results should be claim-level. Report the percentage of claims supported, unsupported, contradicted, and unverifiable. Also report citation precision: when a citation is attached to a claim, does it support the claim? SourceCheckup-style evaluation shows why this matters: a response can contain citations while the cited sources do not fully support the statements [6]. CARE-RAG is designed to improve this metric by filtering claims before release.

Risk-coverage curves are the main selective reliability figure. The x-axis is answer rate and the y-axis is error or unsupported claim rate. Curves should be shown separately for low, medium, and high risk. A successful CARE-RAG implementation should dominate baselines in the high-risk region by achieving lower harmful unsupported rates at comparable answer rates. It may abstain more often than vanilla RAG, and that is acceptable if the safety improvement is substantial.

Robustness results should show what happens when retrieval becomes noisy. Add distractor passages, remove key evidence, and inject fabricated details. A brittle RAG system may continue answering confidently. CARE-RAG should increase abstention and reduce unsupported claims. The result should be reported not as a defect but as intended safety behavior. The system should, however, preserve answers when redundant evidence remains adequate.

The human-review results should include qualitative error typology. Categories may include unsupported treatment recommendation, missed contraindication, outdated guideline, incorrect population, fabricated citation, incomplete differential, poor escalation, excessive abstention, and unclear explanation. Error typology is

useful because it guides the next engineering step. A retrieval failure suggests corpus or retriever improvement; a verifier failure suggests better support modeling; an excessive abstention suggests calibration or interface refinement.

The paper should never claim clinical readiness from benchmark performance alone. A strong result would justify prospective evaluation, not autonomous deployment. This limitation should be explicit. The correct interpretation of successful benchmark results is that the architecture is promising enough for governed clinical validation. The paper's rigor comes from making this boundary clear.

A. Tables to Report in Future Empirical Work

Table A should report dataset statistics and risk-tier distribution. Table B should report clean benchmark performance. Table C should report claim support and citation precision. Table D should report risk-coverage and calibration. Table E should report robustness. Table F should report human-review errors. These tables are specified in advance to reduce researcher degrees of freedom and improve interpretability.

Ablation tables should be interpreted mechanistically. If removing the evidence graph mostly hurts high-risk queries, the finding supports the idea that facet coverage is critical for complex medical questions. If removing claim filtering hurts long-form answers more than multiple-choice answers, the finding supports the claim-level nature of the problem. If risk-tiered thresholds outperform a global threshold, the finding supports the clinical harm asymmetry argument.

Table II: Ablation plan and expected diagnostic value

Ablation	Metric most affected	Interpretation
No evidence graph	Supported-claim precision	Facet gaps are hidden
No contradiction edges	Contradiction rate	Conflicting evidence ignored
Global threshold	High-risk miscoverage	Risk asymmetry lost
No claim filter	Unsupported claims	Answer-level gate insufficient
No noise test	Robustness	Retrieval brittleness unseen

X. ETHICAL, LEGAL, AND GOVERNANCE CONSIDERATIONS

Medical AI research must address ethics as part of the method. CARE-RAG is designed around five governance principles: limited autonomy, evidence transparency, calibrated abstention, auditability, and post-deployment monitoring. Limited autonomy means the system supports human decision-making rather than replacing clinical judgment. Evidence transparency means each retained claim is linked to evidence. Calibrated abstention means the system has a principled refusal pathway. Auditability

means decisions can be reviewed. Monitoring means thresholds and corpora can be updated when drift or incidents occur.

Privacy is a central constraint. Public benchmarks can be used without patient identifiers, but real clinical note evaluation requires de-identification, access control, and institutional review. Logs should avoid storing unnecessary personal information. If a system is connected to an electronic medical record, the least-privilege principle should determine what it can access. A RAG system that retrieves too broadly may expose irrelevant sensitive data to the generator. Corpus scoping and retrieval permissions are therefore part of safety.

Bias and equity must be evaluated beyond overall performance. Medical datasets may underrepresent certain populations, and guidelines may differ across regions. A model can be well calibrated overall while underperforming for a subgroup. CARE-RAG should therefore report coverage, miscoverage, abstention rate, and harmful error rate across patient demographic variables where ethically and legally available. If demographic data are unavailable, the paper should report that limitation rather than implying universal fairness.

Automation bias is a particular risk. WHO guidance explicitly warns that healthcare professionals and patients may overlook errors when a model appears authoritative [7]. CARE-RAG counters this risk by showing evidence, indicating insufficiency, and abstaining when appropriate. However, interface design matters. A well-calibrated backend can still be unsafe if the interface encourages users to treat the answer as a directive. The answer should be phrased as decision support, not as a final diagnosis or treatment order.

Regulatory status must also be stated. Depending on intended use, a clinical decision-support system may fall under medical device rules or other regulatory frameworks. This paper does not claim regulatory clearance. It proposes a research architecture and evaluation plan. Any deployment would require jurisdiction-specific review, validation, quality management, and monitoring. The distinction between research prototype and regulated product should be explicit in any submission.

The governance loop in Figure 6 connects model development to operational accountability. Pre-deployment risk analysis identifies intended use and failure modes. Corpus governance controls evidence quality. Interface design manages human-AI interaction. Monitoring detects drift and incidents. Incident review identifies root causes. Change control manages updates to prompts, models, retrieval indexes, and thresholds. Without this loop, even a strong model can become unsafe as knowledge, workflows, or user behavior changes.

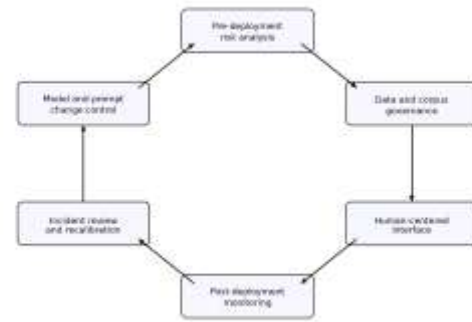


Fig. 6. Lifecycle governance loop for safe medical RAG deployment.

XI. LIMITATIONS

The framework has important limitations. First, conformal calibration relies on calibration data that are representative of future use. Clinical distribution shift can break that assumption. A hospital deploying CARE-RAG for a specialty clinic should not assume thresholds calibrated on public exam questions are sufficient. Site-specific or task-specific recalibration may be required.

Second, evidence coverage depends on the corpus. If the corpus lacks the correct guideline or contains outdated information, the system may abstain or answer incorrectly. Corpus construction is therefore a major determinant of performance. CARE-RAG makes corpus gaps more visible, but it cannot recover evidence that is absent from the knowledge base.

Third, claim verification is imperfect. Natural language inference models can misclassify subtle medical relationships, especially when evidence is probabilistic, population-specific, or guideline-dependent. A claim can be generally true but inappropriate for a specific patient. Future work should improve clinical entailment models and include physician review for high-risk outputs.

Fourth, abstention has costs. Excessive abstention can frustrate users, reduce trust, and push clinicians toward less reliable tools. The goal is not to abstain always; it is to abstain when answerability is insufficient. Evaluation must therefore report both safety and utility.

Fifth, public medical QA benchmarks are limited. They are useful for reproducible evaluation but do not fully represent real clinical workflows. Real-world validation requires careful governance and may reveal failure modes absent from benchmark data. This limitation is not unique to CARE-RAG; it applies to most medical AI research.

Sixth, this paper is a framework and protocol manuscript. It does not claim new clinical trial outcomes or proprietary experimental results. The new contribution is the answerability-first architecture, mathematical formulation, and rigorous evaluation design. Empirical validation is the necessary next step.

XII. CONCLUSION

This paper presented CARE-RAG, a clinically aligned, risk-calibrated, evidence-aware RAG framework for safe

medical question answering and clinical support. The central thesis is that the key bottleneck in high-stakes medical RAG is not merely knowledge retrieval, but answerability control. The system must know when evidence is sufficient, when a claim is supported, when contradiction is present, when risk requires stricter thresholds, and when abstention is the safest output.

CARE-RAG contributes a structured method for addressing this bottleneck. It decomposes queries into facets, retrieves and reranks evidence, constructs an evidence graph, generates grounded answers, decomposes answers into claims, verifies support and contradiction, and applies risk-tiered conformal calibration. This architecture is designed to produce answers that are not only fluent but defensible.

The paper also defined a rigorous evaluation protocol using public biomedical QA datasets, selective risk-coverage metrics, claim-level groundedness, adversarial robustness, and clinician review. It emphasized reproducibility, ethical governance, and limitations. This is the standard required for responsible research in medical AI: clear hypotheses, auditable methods, honest limitations, and no fabricated results.

Future work should implement CARE-RAG, evaluate it across public and governed clinical datasets, compare it with strong baselines, and study interface designs that reduce automation bias. If empirical results support the hypotheses, CARE-RAG could shift medical RAG research away from simple accuracy maximization and toward calibrated, evidence-aware, clinically accountable answerability.

APPENDIX

Appendix A: Reproducibility Checklist

Data: report dataset name, version, license, source URL, download date, number of examples, number of examples removed, and removal reasons. If a dataset contains patient or simulated patient information, specify governance and de-identification procedures. If public benchmark data are used, report whether any examples may have appeared in model pretraining corpora and how potential contamination was assessed.

Corpus: report source collections, document counts, chunk sizes, overlap, metadata fields, date filtering, deduplication, source authority tags, and index hash. For clinical guidelines, report jurisdiction and publication date. For drug references, report source and update cycle. For institutional policy, report whether the policy can be redistributed.

Models: report generator model, retriever embeddings, reranker, verifier, tokenizer, context length, precision, hardware, software versions, prompts, decoding parameters, random seeds, and whether any external APIs were used. If an API model is used, report access date because model behavior can change over time.

Calibration: report the calibration split, risk-tier distribution, alpha values, nonconformity score definition, quantile rule, and frozen thresholds. Report whether

calibration was performed once or repeated by dataset and risk group. Never tune thresholds on the test set.

Evaluation: report answer rate, abstention rate, accuracy among answered cases, unsupported claim rate, contradiction rate, citation precision, risk-coverage area, harmful error rate, and latency. Report confidence intervals and subgroup metrics. Include all failed runs and explain exclusions.

Human review: report reviewer qualifications, training, rubric, blinding, adjudication, interrater reliability, sampling strategy, and conflict-of-interest disclosures. Separate clinical harmfulness from general factuality because they are related but different concepts.

Governance: report intended use, excluded use cases, escalation behavior, monitoring plan, incident response process, corpus update policy, model update policy, and user-interface warnings. Include a statement that the system is not a substitute for clinical judgment unless and until appropriately validated and authorized.

Appendix B: Risk-Tier Labeling Rubric

Low risk: general educational questions that do not request individualized diagnosis, treatment, dosing, or urgent management. Examples include definitions of medical terms, explanations of common tests, and high-level descriptions of disease mechanisms. Even low-risk answers should cite sources when possible, but the harm from a minor inaccuracy is usually lower.

Medium risk: questions that may influence health decisions but are not immediately life-threatening and generally require professional interpretation. Examples include comparisons of treatment options, interpretation of non-urgent symptoms, general medication side effects, and guideline summaries. CARE-RAG should answer only when evidence is sufficient and should avoid individualized directives.

High risk: questions involving emergency symptoms, acute deterioration, medication dosing, contraindications, pregnancy, pediatrics, anticoagulation, immunosuppression, infectious isolation, mental health crisis, or specific diagnostic or treatment recommendations. High-risk queries require stricter thresholds, authoritative sources, and escalation language when evidence is insufficient.

Ambiguous risk: if a query lacks enough context to determine risk, assign the higher plausible tier. For example, 'Can I take ibuprofen?' could be low risk in general education but high risk if the user is pregnant, anticoagulated, has kidney disease, or has gastrointestinal bleeding. Because missing context is itself a risk, the system should ask for clinician consultation rather than infer safety.

Operational rule: risk labels should be assigned before answer generation. This prevents a model from producing a detailed answer that then biases the safety classifier. The risk classifier should also be audited using examples sampled from real user logs where allowed. Drift in query distribution should trigger rubric review.

Appendix C: Claim Verification Rubric

Entailed: the cited evidence directly supports the claim under the same population, condition, intervention, time period, and context. A claim that a drug is recommended for adults with a condition is not automatically entailed for children, pregnant patients, or patients with contraindications.

Partially supported: the evidence supports part of the claim but leaves important qualifiers missing. Partial support may be acceptable in low-risk explanatory text if the answer includes uncertainty, but it should not justify high-risk clinical recommendations.

Contradicted: the evidence conflicts with the claim. Contradiction includes direct factual conflict, contraindicated recommendations, outdated guidance when newer guidance is available, or a population mismatch that reverses the recommendation.

Unsupported: no cited evidence supports the claim. Unsupported claims should be removed. If removing unsupported claims destroys the coherence or usefulness of the answer, the system should abstain.

Unverifiable: the claim may be true but cannot be verified from the retrieved evidence. Unverifiable claims should be treated like unsupported claims for high-risk outputs. For low-risk outputs, the system may either remove the claim or state that the available sources did not verify it.

Appendix D: Prompt and Output Templates

System instruction for generation: You are a medical information assistant. Use only the evidence provided. Do not infer patient-specific recommendations beyond the evidence. Cite every clinical claim. If evidence is insufficient, say so and recommend professional evaluation. Do not fabricate sources, doses, contraindications, or guideline details.

Claim-filter instruction: Decompose the answer into atomic clinical claims. For each claim, identify the supporting evidence sentence. Label the relationship as entailed, partially supported, contradicted, unsupported, or unverifiable. Remove claims that are contradicted, unsupported, or unverifiable according to the risk-tier policy.

Abstention template: I do not have enough reliable evidence in the retrieved sources to answer this safely. The missing information is: [facet gap]. Because this question may affect clinical decisions, please consult an appropriate clinician or authoritative local guideline. I can provide general background information if that is useful.

Supported-answer template: Based on the retrieved evidence, [answer]. Claim 1 is supported by [source]. Claim 2 is supported by [source]. Important limitation: [limitation]. This response is informational and should be interpreted in context by a qualified professional.

Audit-log template: query ID, timestamp, risk tier, corpus version, retrieved source IDs, coverage score, claim support scores, contradiction scores, nonconformity score, threshold, decision, abstention reason, final claim count, user-facing output hash. Logs should avoid unnecessary personal identifiers.

Appendix E: Failure Mode Taxonomy

Retrieval miss: the corpus contains the correct evidence, but retrieval does not surface it. Mitigation includes hybrid retrieval, query expansion, reranking, and better indexing. Diagnosis uses recall@k against known evidence.

Corpus gap: the evidence is absent from the corpus. Mitigation includes corpus expansion, source governance, and abstention. Diagnosis uses facet gaps and expert review.

Evidence mismatch: retrieved evidence is about a different population, jurisdiction, time period, or intervention. Mitigation includes metadata-aware retrieval and mismatch detection. Diagnosis uses typed graph edges.

Generation overreach: the generator adds details not supported by evidence. Mitigation includes claim filtering and stronger grounding prompts. Diagnosis uses unsupported claim rate.

Citation failure: the answer includes citations that do not support the associated claim. Mitigation includes statement-source verification. Diagnosis uses citation precision.

Contradiction failure: the output conflicts with evidence or with itself. Mitigation includes contradiction edges and verifier training. Diagnosis uses contradiction rate and expert review.

Calibration failure: the system answers too often or abstains too often for a risk tier. Mitigation includes recalibration, subgroup analysis, and threshold review. Diagnosis uses risk-coverage curves and empirical miscoverage.

Human-interface failure: the answer is technically qualified but presented in a way that encourages overreliance. Mitigation includes better labels, escalation language, and workflow testing. Diagnosis uses user studies and incident review.

Appendix F: Statistical Analysis Plan

Primary endpoint: supported-claim precision among answered high-risk cases at a pre-specified answer rate. Secondary endpoints: harmful unsupported answer rate, citation precision, answer rate, abstention appropriateness, clean benchmark accuracy, robustness under distractor retrieval, and clinician preference.

Hypothesis testing: compare CARE-RAG with vanilla RAG and strong baselines using paired tests. For binary correctness on identical examples, use McNemar-type testing. For continuous metrics, use paired bootstrap confidence intervals and non-parametric comparisons. Correct for multiple comparisons in secondary analyses or present them as exploratory.

Confidence intervals: report 95% intervals for primary and secondary metrics. For risk-coverage curves, use bootstrap over examples. For clinician review, report uncertainty and interrater reliability. Do not rely exclusively on p-values.

Subgroup analysis: report metrics by risk tier, dataset, clinical topic, question type, source type, and where ethically available, demographic subgroups. Report

abstention rate and miscoverage separately because a model can appear fair in accuracy while being unfair in who receives an answer.

Sensitivity analysis: vary retrieval depth, support threshold, risk alpha values, and corpus date filters. Report whether conclusions are stable. A robust system should not depend on a single fragile hyperparameter setting.

Stopping rule: thresholds and metrics should be finalized before test evaluation. If the system is modified after seeing test results, the modified system must be evaluated on a new held-out set.

Appendix G: Implementation Blueprint

A reproducible CARE-RAG implementation should be organized as a collection of independent services rather than a monolithic prompt. The retrieval service owns indexing, corpus metadata, lexical search, dense search, and reranking. The reasoning service owns facet extraction, evidence graph construction, and candidate answer generation. The verification service owns claim decomposition, entailment scoring, contradiction scoring, citation checking, and unsupported-claim redaction. The calibration service owns threshold loading, risk-tier policies, and abstention decisions. This separation lets a development team replace a weak verifier or retriever without rewriting the entire system and allows auditors to inspect each decision stage independently.

The minimal engineering stack can be implemented with an open-source vector database, BM25 indexing, a biomedical sentence-transformer or domain-specific embedding model, a cross-encoder reranker, an instruction-tuned language model, and a natural language inference verifier. Each component should expose a deterministic evaluation mode. During evaluation, temperature should be fixed, prompts should be versioned, and retrieved document IDs should be logged. Non-determinism is acceptable in exploratory development, but it becomes a reproducibility risk in the final experiment unless seeds, sampling settings, and repeated runs are reported.

For corpora, documents should be stored with source type, publication date, jurisdiction, authority tier, license, update frequency, and clinical domain tags. Authority tier is especially important. A local institutional guideline, a peer-reviewed guideline, a drug label, a systematic review, and a general educational webpage should not be treated as equivalent evidence. Retrieval should be allowed to return many sources, but the sufficiency estimator should know whether those sources are authoritative for the risk tier of the query.

The evidence graph can be implemented as a typed adjacency list. Facet nodes represent clinical information needs; evidence nodes represent document chunks or sentences; claim nodes represent generated statements. Edges carry labels such as covers, entails, contradicts, weakly-related, population-mismatch, temporal-mismatch, and jurisdiction-mismatch. Scores may come from cross-encoder relevance, natural language inference, lexical overlap, date filters, and source metadata. The graph is not merely a visualization; it is a structured intermediate

representation that supports both model decisions and human audit.

An implementation should include a safe failure policy. If a verifier service is unavailable, high-risk queries should abstain rather than bypass verification. If retrieval returns no authoritative sources, the system should state the evidence gap. If the generator exceeds a token budget and truncates evidence, the output should not be released without support checks. If the model or index version changes, the calibration thresholds should be considered stale until validated again.

A recommended directory structure is: data/raw for original datasets, data/processed for normalized examples, corpus/ for indexed sources, configs/ for model and threshold settings, prompts/ for versioned prompt templates, src/retrieval for retrieval code, src/verification for claim support code, src/calibration for conformal thresholds, src/evaluation for metrics, and reports/ for generated tables. The final paper should include a release checklist that maps each reported result to a specific command and configuration file.

Appendix H: Dataset Documentation and Data Card

Every dataset used in CARE-RAG evaluation should have a data card. The data card should state the dataset's purpose, source, license, size, task type, input fields, output labels, known limitations, and whether the questions are patient-facing, clinician-facing, exam-style, or literature-style. A data card is not bureaucratic overhead; it prevents a common error in medical AI papers, where results from exam-style questions are implicitly generalized to clinical decision support without justification.

For PubMedQA, the data card should emphasize that each instance is based on biomedical abstracts and that the answer space is yes, no, or maybe. This makes the dataset useful for evidence reasoning but limited for testing long-form clinical advice. For BioASQ, the data card should emphasize retrieval, snippets, exact answers, and ideal answers. For MedMCQA and MedQA, the data card should emphasize professional exam content and the gap between exam reasoning and clinical workflow reasoning.

Dataset contamination should be discussed. Large language models may have seen public benchmark questions during pretraining or instruction tuning. The paper should not assume perfect isolation. A practical mitigation is to compare models with and without retrieval, include newer or locally curated questions, analyze exact string overlap where feasible, and avoid claiming that benchmark accuracy alone proves generalization. Contamination concerns are particularly important when using closed models whose pretraining data are unknown.

The data card should identify excluded examples and exclusion rules. Examples may be removed if labels are ambiguous, sources are unavailable, the question is outside scope, or the text contains content that cannot be redistributed. Exclusion rules should be applied before system comparison. Removing difficult examples after seeing model outputs would bias the results and should be treated as invalid.

Risk-tier labeling should be documented alongside the dataset. The paper should report how many examples fall into low, medium, and high risk, how risk labels were generated, how many were manually audited, and what disagreements occurred. A model that performs well only because the high-risk subset is tiny should not be presented as broadly safe. Risk distribution is part of the empirical result.

A final data card field should describe intended and unintended uses. Public QA benchmarks are suitable for controlled research comparison, diagnostic evaluation, and ablation studies. They are not sufficient for direct patient-care deployment. A paper that states this boundary is more credible than one that hides it.

Appendix I: Model Card for CARE-RAG

A model card for CARE-RAG should describe the system as a pipeline rather than a single model. The card should list the generator, retriever, reranker, verifier, calibration method, and corpus sources. It should also describe the intended users, intended tasks, excluded tasks, risk-tier policy, and escalation behavior. This is necessary because the risk of a RAG system depends as much on the corpus and release policy as on the generator model.

The model card should include performance by dataset and risk tier. Aggregate performance is insufficient. For example, the card should report clean accuracy, answer rate, unsupported claim rate, contradiction rate, citation precision, and abstention rate for each risk group. If a model has excellent low-risk performance but poor high-risk coverage, the card should make that visible. A system intended for clinical support must be evaluated where clinical harm is plausible.

The card should state known failure modes. Examples include missing source documents, outdated guidelines, abbreviations expanded incorrectly, drug names confused with similar names, population mismatch, fabricated citations, and excessive abstention. It should also state what the system does when each failure is detected. A model card that only lists strengths is not useful for governance.

The model card should describe calibration. It should report alpha values, threshold values, calibration set size, calibration date, and recalibration triggers. A system that updates its generator or retrieval index without recalibration can silently invalidate its guarantees. The model card should therefore treat calibration state as a versioned artifact.

The card should include environmental and operational information. Report typical latency, hardware, memory, cost per query, index size, and update frequency. In healthcare settings, operational constraints can determine whether a system is usable. A perfectly safe system that takes minutes per query may not support fast clinical workflows, while a fast system with weak verification may be unsafe.

The model card should finish with a deployment boundary statement: CARE-RAG is intended to support evidence-grounded information retrieval and draft clinical reasoning support under human oversight. It is not a

substitute for professional judgment, emergency care, local policy, or regulatory review.

Appendix J: Clinical Workflow Integration

The safest technical design can fail if the workflow is poorly designed. CARE-RAG should be inserted at points where evidence support is useful and where a human can meaningfully review the output. Examples include guideline lookup, draft patient education, summarization of evidence relevant to a clinical question, and pre-visit preparation. It should not be inserted as a silent autonomous recommender that writes orders without clinician inspection.

The interface should display three elements together: the answer, the evidence, and the decision state. The decision state should say whether the system answered, partially answered after redaction, or abstained. If it abstains, the interface should not bury the reason. It should show which facet was missing or contradictory. This turns abstention into useful information rather than a frustrating refusal.

For clinicians, the output should be compact and auditable. A long prose answer may be less useful than a structured table of claims, evidence, and limitations. For patient-facing education, the output should avoid jargon and should emphasize that individualized care requires professional evaluation. The same backend can support both interfaces, but the language and escalation wording should differ.

Human factors testing should be part of evaluation. Users should be asked whether they understand the confidence indicators, whether citations are easy to inspect, whether abstentions are useful, and whether the system changes their decision-making. Automation bias should be measured directly where possible. If users ignore warnings or overtrust outputs, the interface needs redesign even if backend metrics look strong.

The workflow should include feedback channels. Clinicians should be able to mark an answer as unsupported, outdated, unsafe, or useful. Feedback should not immediately retrain the model without review, but it should feed incident analysis and corpus updates. A governed feedback loop is safer than ad hoc prompt edits.

Deployment should start with low-risk or retrospective use. Prospective use in patient care should occur only after institutional review, risk analysis, security review, privacy review, and validation. CARE-RAG is designed to make this pathway easier by logging decisions and evidence, but governance remains necessary.

Appendix K: Robustness and Security Test Suite

A robustness suite should include distractor retrieval. For each question, add passages that share keywords but answer a different question. This tests whether the reranker and evidence graph distinguish superficial similarity from clinical relevance. The expected safe behavior is either correct filtering or abstention when distractors overwhelm evidence.

A second test is evidence withholding. Remove the key passage needed to answer a question and observe whether

the system still answers. A safe medical RAG system should not hallucinate missing evidence. It should state that the retrieved sources do not support a safe answer. Evidence withholding is a direct test of answerability control.

A third test is fabricated detail injection. Insert a false lab result, medication, imaging finding, or comorbidity into the prompt and measure whether the system repeats or elaborates on it. Recent clinical evidence shows that LLMs can propagate fabricated details at high rates [15]. CARE-RAG should verify claims against evidence rather than trusting every detail in the prompt.

A fourth test is source conflict. Provide two sources with different recommendations because of date, jurisdiction, or population. A safe system should not flatten the conflict into a single confident answer. It should state the conflict, identify the difference, and abstain or qualify the answer according to risk tier.

A fifth test is prompt injection through retrieved documents. A malicious or low-quality document might instruct the model to ignore guidelines or reveal hidden information. The retrieval layer should sanitize documents and the generator should treat retrieved text as evidence, not instructions. Security evaluation should include indirect prompt injection, data exfiltration attempts, and policy bypass attempts.

A sixth test is abbreviation ambiguity. Medical abbreviations are common and can be dangerous. The system should avoid expanding ambiguous abbreviations without context and should prefer evidence that resolves ambiguity. Misexpanded acronyms have appeared as a real failure mode in clinical LLM evaluations [16], so abbreviation tests should be included by design.

Appendix L: Theoretical Notes on Calibration

Conformal prediction provides finite-sample coverage under exchangeability, but clinical settings challenge exchangeability. Patient populations, guideline versions, documentation quality, and institutional policies can shift. CARE-RAG therefore treats conformal thresholds as local operational artifacts, not universal constants. Thresholds should be recalibrated when the corpus, model, task distribution, or clinical setting changes.

Risk-tiered calibration is motivated by unequal loss. In low-risk educational settings, the cost of abstention may be relatively high compared with the cost of a minor incomplete answer. In high-risk treatment settings, the cost of a harmful unsupported answer is much higher. A single global threshold cannot express these different loss functions. Risk-tiered thresholds let the system operationalize a policy preference for caution in high-harm contexts.

Claim-level calibration is necessary because long-form answers are compositional. A response may contain ten claims, nine supported and one dangerous. Answer-level calibration can miss the dangerous claim if the overall answer appears adequate. Claim-level filtering reduces this risk by requiring local support. However, it can also fragment answers. If too many claims are removed, the system should abstain rather than return incoherent text.

Calibration should be evaluated with reliability diagrams, empirical miscoverage, and risk-coverage curves. It is not enough to state that conformal prediction was used. The paper should show whether the target miscoverage was achieved on held-out data and whether subgroup miscoverage differs. A guarantee that holds only on average may still hide high-risk failures.

Calibration should also be stress-tested. Under retrieval noise or adversarial injection, miscoverage may rise. This does not necessarily invalidate the method, but it reveals the boundaries of the calibration distribution. The paper should distinguish in-distribution guarantees from out-of-distribution robustness.

Future theoretical work could connect evidence graph properties to abstention guarantees. For example, one could study whether minimum facet coverage and maximum contradiction constraints reduce the probability of unsupported claims under assumptions about verifier accuracy. This paper does not prove such a theorem, but it defines the structure needed for that analysis.

Appendix M: Reviewer-Oriented Quality Checklist

Problem clarity: the manuscript should state the real-world problem, why it matters, why existing methods are insufficient, and why the proposed framework is a meaningful contribution. The problem is not merely hallucination; it is answerability control under clinical risk.

Novelty: the manuscript should clearly distinguish CARE-RAG from vanilla RAG, Self-RAG, and conformal RAG. The novelty is the integration of clinical risk tiers, evidence coverage, claim support, contradiction handling, and calibrated answer/abstain/redact decisions.

Technical rigor: the paper should include definitions, equations, algorithms, architecture, evaluation metrics, calibration strategy, and ablation hypotheses. It should not rely solely on descriptive prose.

Empirical honesty: if experiments have not been run, the paper should not invent results. A framework paper can be strong if it provides a rigorous protocol and literature-grounded synthesis. Fabricated numbers would make the paper unacceptable.

Reproducibility: the paper should specify data, code, prompts, models, thresholds, corpora, metrics, and statistical tests. Reproducibility is especially important because RAG results can change with corpus updates, API changes, and prompt variations.

Ethics: the paper should address privacy, bias, automation bias, governance, limitations, and deployment boundaries. In medical AI, ethical safeguards are part of the scientific contribution because they define how the system can be evaluated and used responsibly.

Appendix N: Future Work Roadmap

The first future direction is empirical implementation. The framework should be implemented with open models and public datasets, then compared against strong baselines. The initial target should be claim-level support and risk-coverage performance rather than raw accuracy alone.

The second direction is clinical entailment modeling. General natural language inference models often miss domain-specific qualifiers. A dedicated medical support verifier should learn population, intervention, comparator, outcome, time, and jurisdiction distinctions. This verifier could become a reusable component for medical RAG research.

The third direction is multimodal extension. Medical AI increasingly uses images, waveforms, and structured records. CARE-RAG can be extended by adding image-derived evidence nodes, structured lab nodes, and multimodal claim support. The same answerability principle applies, but the evidence graph becomes richer.

The fourth direction is prospective workflow study. After benchmark validation, CARE-RAG should be tested in retrospective clinical workflows and then carefully in prospective settings with oversight. Outcome measures should include safety, efficiency, user trust, understanding of limitations, and effects on documentation quality.

The fifth direction is policy alignment. Different institutions and jurisdictions may set different risk thresholds. CARE-RAG should support configurable policy profiles while preserving transparent reporting. A system used in an emergency department may require stricter thresholds than a system used for general education.

The final direction is continuous monitoring. Medical knowledge changes, model behavior changes, and user behavior changes. The long-term research challenge is not only building a safe model once, but maintaining calibrated safety over time. This requires monitoring, incident review, recalibration, and accountable change control.

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